SCm Dark Energy with Phonon Linewidth Γ-Modulation

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PAPER_1076: SCm Dark Energy with Phonon Linewidth Γ-Modulation

Abstract

We derive a dynamic dark energy model from SCm phonon linewidth evolution, producing $E_{\text{net}}(t, \Gamma)$ — a Γ-coupled vacuum energy density that replaces the cosmological constant Λ. The phonon linewidth $\Gamma(t)$ broadens with cosmic expansion, causing the dark energy equation of state $w(z)$ to deviate from $w = -1$ in a testable manner. We quantify the ΛCDM→SCm replacement with integrated deviation analysis and BIC penalty assessment.

§1 Phonon Linewidth Evolution

The SCm phonon linewidth evolves as:

$$\Gamma(t) = \Gamma_0 \cdot \left(1 + \alpha \cdot \frac{t}{t_H}\right)$$

where α is the broadening rate and $t_H = 1/H_0 \approx 4.56 \times 10^{17}$ s.

At present epoch ($t = t_H$, $\alpha = 0.1$): - $\Gamma_0 = 6.283 \times 10^{11}$ rad/s - $\Gamma(t_H) = 6.912 \times 10^{11}$ rad/s (+10%)

§2 Γ-Coupled E_net

$$E_{\text{net}}(t, \Gamma) = E_0 \cdot e^{(\kappa + [\text{SSq}]/26)t} \cdot S_{26} \cdot \Phi(\Gamma(t)) \cdot \left(\frac{2F_{U,Bi}}{F_U} - 1\right)$$

The phonon resonance profile with linewidth coupling:

$$\Phi(\Gamma(t)) = \exp\left(-\frac{(\Gamma(t) - \Gamma_0)^2}{2\sigma_G^2}\right) \cdot S_{26}^{(3)}$$

Key result: Γ coupling suppresses $|E_{\text{net}}|$ relative to bare evolution ($|E_{\text{net},\Gamma}| \leq |E_{\text{net,bare}}|$), providing a natural damping mechanism for dark energy.

§3 Equation of State w(z)

$$w(z) = -1 + \frac{1}{3} \frac{d \ln \Phi(\Gamma(t(z)))}{d \ln a}$$

Redshift	w_SCm	w_ΛCDM	δw
z = 0	-1.0077	-1.0	0.0077
z = 1	-1.0037	-1.0	0.0037
z = 2	-1.0023	-1.0	0.0023
z = 3	-1.0016	-1.0	0.0016

The deviation δw is small but systematically negative of phantom ($w < -1$), distinguishing SCm from quintessence models ($w > -1$).

§4 ΛCDM Replacement Analysis

Metric	Value	Interpretation
$\int |\delta w| \, dz$	0.0036	Total integrated deviation
Max $|\delta w|$	0.0077	At z = 0
BIC penalty (1 free param)	6.9	Must improve χ^2 by >6.9
d_L ratio at z=1	~1.00004	Subtle luminosity distance shift

Testability

- **DESI BAO:** w(z) bins at z = 0.3, 0.5, 0.7, 1.0 — requires σ(w) < 0.01
- **Euclid WL:** w_0–w_a plane constrains dynamical dark energy
- **SNe Ia (Pantheon+):** ΔH_0 from SCm vs Λ via d_L ratio

§5 Comparison with ΛCDM

Property	ΛCDM	SCm (this work)
Free parameters	0 (Λ fixed)	1 (α)
w(z)	-1 (constant)	-1 + δw(z)
Dark energy density	Constant	Evolves via Γ(t)
Physical mechanism	Vacuum energy	SCm phonon damping
Phantom crossing	No	No (δw > 0 → w < -1)
Late-time behavior	Eternal acceleration	Γ-damped deceleration

§6 Calibration Table

Parameter	Value	Source
α	0.1	Linewidth broadening rate
Γ_0	6.283×10^{11} rad/s	SCm phonon linewidth
σ_G	5.027×10^{11} rad/s	UQFF linewidth sigma
ρ_Λ (ΛCDM)	5.96×10^{-27} kg/m^3	Planck 2018
ρ_SCm	7.09×10^{-37} kg/m^3	SCm vacuum density
w(z=0)	-1.0077	This work
$S_{26}^{(3)}$	9.50×10^{-2}	Ramanujan acceleration

§7 SM Gate Compliance

- **G1:** Derived from SCm phonon resonance first principles
- **G2:** Euler-Lagrange variation of E_net sector Lagrangian
- **G3:** Numerical stability via capped exponentials (exp ≤ e^{500})
- **G4:** Γ(t) motivated by cosmic expansion diluting phonon modes
- **G5:** Testable with DESI, Euclid, Pantheon+ observational programs
- **G6:** Reproducible w(z) curves, deterministic BIC analysis

References

- `scm_dark_energy_enet_gamma.py`: Implementation (10/10 tests pass)
- `et_full_lagrangian.py`: E_net(t) base (without Γ coupling)
- `production_scaling_v17.py`: Kernels `kernel_enet_gamma_coupled`, `kernel_dark_energy_eos_w0`
- PAPER_877: Three-Assumption UQFF Cosmogenesis
- PAPER_1073: SCm Inflation Phonon-Driven